

Technical Document #913: Retrieval Augmented Generation - Comprehensive Overview

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Hallucination detection identifies when LLMs generate factually incorrect information. It is critical for maintaining trust in RAG systems.

Query decomposition breaks complex queries into simpler sub-queries. It improves retrieval quality for multi-faceted questions.

Context window optimization selects the most relevant information to fit within the LLM's context limit. It maximizes information density.

Hybrid search combines keyword-based and semantic search. It leverages the strengths of both approaches for better retrieval quality.

The rapid advancement of technology has transformed how organizations approach problem-solving and innovation. Companies must adapt to stay competitive in an ever-evolving landscape.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Key Takeaways:

- Context window optimization selects the most relevant information to fit within the LLM's context limit.
- Query decomposition breaks complex queries into simpler sub-queries.
- Evaluation metrics for RAG include faithfulness, answer relevance, and context recall.